

BELIMO

Energy Valve Coil Estimator

(Belimo Energy Valve Coil Estimator.xlsm)

USER MANUAL

Release 4.5

Prepared for Belimo internal staff – not for distribution.

Belimo U.S., Inc. | 33 Turner Road, Danbury, CT 06810

Languages: English · Español · Português · Français | Units: US or Metric

Contents

1. Introduction	4
1.1 How the workbook is laid out	4
1.2 What's in this manual	4
2. Getting Started	6
2.1 Opening the file	6
2.2 The About screen	6
2.3 The step callouts	7
2.4 Which cells can I type in?	7
2.5 Language and units	7
3. The Home Tab – the Five Steps	9
3.1 Steps 1–2: Language and coil design data	9
3.2 Steps 3–4: Operating time and climate region	10
3.3 Step 5: Financial inputs and results	11
3.4 The help icons and the Disclaimer	12
4. The Pump Savings Tab	13
4.1 Design data	13
4.2 Energy Valve selection	13
4.3 The performance blocks	14
4.4 Saving the tab as a PDF	14
5. Flow Curves and Charts	16
5.1 Flow Curves	16
5.2 Charts	16
6. The Results Tab	18
6.1 What each block shows	18
6.2 Greenhouse gas equivalents and financial summary	19
6.3 Utility rebate support	19
7. The Financial Results	21

7.1 First, the idea behind all three: present value	21
7.2 Net Present Value (NPV)	21
7.3 Savings-Investment Ratio (SIR)	21
7.4 Simple Payback Period	21
7.5 Reading the four horizons	22
7.6 Worked example (the values the workbook opens with)	22
8. How the Numbers Are Calculated	23
9. Practicalities and Troubleshooting	24
9.1 Presenting and printing	24
9.2 Language	24
9.3 If something looks wrong	24
Appendix. Release History	25

1. Introduction

The Belimo Energy Valve Coil Estimator (the “tool”, file name Belimo Energy Valve Coil Estimator.xlsm) is an Excel workbook that answers one question about one air handler: *“So... how much will this smart valve save me on this coil?”* You enter the coil’s design data, how long it runs, where the building is, and the Delta-T the coil delivers today and the Delta-T an Energy Valve will hold. The tool works out how much less water the pump has to move, what that saves in energy and money, and what the investment is worth over 5, 10, 15 and 20 years.

More flow costs money, and it costs it disproportionately: pump power varies with roughly the cube of flow, so cutting flow by a third cuts pump power by around 70%. An Energy Valve holds the coil at the Delta-T you ask for, so the flow – and the pumping bill – comes down.

Scope of the tool

This is a **coil-level** tool and it counts **pumping savings only**. Its companion, the Energy Valve Savings Estimator, models a whole chilled-water plant and adds the chiller-side savings. Use this tool when the conversation is about a specific coil or air handler; use the plant tool when it is about the central plant. It is an estimating tool for internal use. Results depend on the accuracy of the data entered and do not take external factors into account.

1.1 How the workbook is laid out

Tab	What it is for
Home	Everything you fill in, in five numbered steps, plus the headline financial results.
Pump Savings	The calculation engine: one block per Delta-T, plus Energy Valve model selection.
Flow Curves	Reference curves relating the control signal to flow through the valve.
Charts	The same relationships plotted.
Results	The customer-facing summary. This is the page you present.

1.2 What’s in this manual

- Section 2 – Getting started: opening the file, the About screen, updates, which cells you can type in, language and units.
- Section 3 – The Home tab, step by step.
- Section 4 – The Pump Savings tab.
- Section 5 – Flow Curves and Charts.
- Section 6 – The Results tab.

- Section 7 – The financial results explained.
- Section 8 – How the numbers are calculated (for anyone who wants to check the arithmetic).
- Section 9 – Practicalities and troubleshooting. Appendix – Release history.

2. Getting Started

2.1 Opening the file

Open the workbook in Excel for Windows. It contains macros, so if Excel shows a yellow **Protected View** bar, click **Enable Editing**, and if it then shows a **Security Warning** bar, click **Enable Content**. (For a file that came from e-mail or a download, right-click it in File Explorer, choose Properties, tick **Unblock** and click OK first.)

On opening, the tool checks Belimo's update source for a newer release (the first time a new one appears it offers to install it), takes you to the Home tab, and re-applies your saved language. The workbook opens with a worked example already filled in (a 310 GPM coil in the North East) so you can see every result before you enter your own data. Overwrite the example values in Steps 2 to 5 with your own.

2.2 The About screen

Click the small round **(i)** button at the top right of the Home tab (beside the Step 1 callout) to open the About screen. It shows the tool's name, version and release date, a short description, and who maintains the tool.

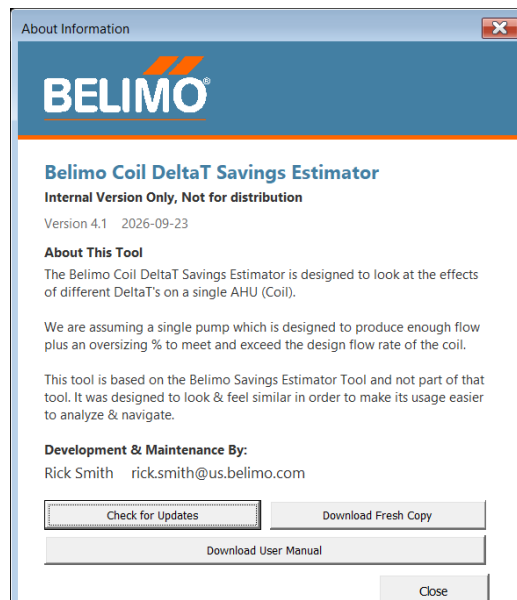


Figure 1 – The About screen.

Button	What it does
Check for Updates	Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.
Download Fresh Copy	Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged.

Button	What it does
Download User Manual	Saves the current published copy of this manual (a PDF) to a folder you choose. If none has been published yet, the tool tells you so.
Close	Closes the About screen.

When a newer release is published the tool tells you the next time you open it, once for each new version; Check for Updates on the About screen does the same at any time and offers a summary of what changed. If you choose to install it, you pick where to save the new copy through a normal Windows **Save As** dialog. Your current copy is never changed or deleted. If the update source cannot be reached, the check does nothing and the tool works as normal.

2.3 The step callouts

The Home tab is laid out as a wizard. Each orange section band has a numbered blue **Step** callout to its right telling you what to enter in that block. Work top to bottom and the tool takes care of the rest.

2.4 Which cells can I type in?

Shading	Meaning
Pale / near-white	Editable. Type here. (The key “Editable Field” / “Label” is shown on the sheets.)
Blue-grey	A label or a calculated result. Protected – the tool writes it.
Orange band	A section heading.

The sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result rather than an input.

2.5 Language and units

Control	Cell	Choices and effect
Language	F5	English, Español, Português or Français. Translates every label, heading, unit word and help message on every tab.
Units	F6	US or Metric. Switches every flow, head, temperature, power and volume figure and its unit.

The calculation always runs in US units internally and converts for display, so switching back and forth never drifts. Figure 2 shows the same Home tab in Spanish with Metric units. Note that the values you have already entered are converted, not re-typed: 310 GPM becomes 1173.478 L/min and a 12 °F Delta-T becomes 6.667 K (shown as °C).



Belimo Energy Valve
Estimador de Ahorros ΔT de Serpentina



Selección de Idioma y Unidades

Idioma

Unidades

Español

Metric

Datos de Entrada de Diseño de la Serpentina

	Valor	Unidad
Costo de Electricidad (Promedio)	0.10	\$/kWh
Carga de Diseño de la Serpentina	1173.478	LPM
Número de Serpentinass	1	uds.
Horas de Operación / Día	20	horas
Días de Operación / Semana	6	días
Semanas de Operación / Año	52	semanas
Altura Real de la Bomba Primaria (Diseño)	89.672	kPa
Eficiencia de Bomba, VFD, Motor (Promedio)	70	%
Delta T de Diseño	6.667	°C
Delta T Real (Antes de la Instalación de Energy Valve)	3.333	°C
Delta T Futuro (Ajuste del Gestor de Delta T)	6.667	°C

Paso 1

Selecione su Idioma y Unidades.

Paso 2

Ingrese en los recuadros de la izquierda, lo siguiente:
Costos Eléctricos
Caudal de Diseño de la Serpentina
Horas de Operación
Altura de la Bomba
Eficiencia del Motor
Valores de Delta T

Figure 2 – The top of the Home tab in Spanish with Metric units.

3. The Home Tab – the Five Steps

Everything you type lives on the Home tab. Figure 3 shows Steps 1 and 2.

3.1 Steps 1–2: Language and coil design data

Field	Value	Unit
Electricity Cost (Average)	0.10	\$/kWh
Design Coil Load	310	GPM
Number Of Coils	1	pcs.
Operating Hours / Day	20	hours
Operating Days / Week	6	days
Operating Weeks / Year	52	weeks
Actual Primary Pump Head (Design)	30	ft.
Pump, VFD, Motor Efficiency (Average)	70	%
Design Delta T	12	°F
Actual Delta T (Prior To Energy Valve Installation)	6	°F
Future Delta T (Energy Valve Delta T Manager Setting)	12	°F

Figure 3 – Steps 1 and 2 – language and units, and the coil design data (with the example values).

Step 1 sets language and units (Section 2.5). Step 2 is the coil itself – the values every later number is built from.

Field	Cell	Unit	What it is
Electricity Cost (Average)	F10	\$/kWh	Blended all-in rate. Every dollar figure in the workbook runs through this one number.
Design Coil Load	F11	GPM	The coil's design flow . Despite the word "Load", this is a flow rate in gallons per minute (or L/min in Metric), not a tonnage – note the unit beside it.
Number Of Coils	F12	pcs.	Fixed at 1 – this tool estimates one coil at a time. (Its help icon notes multiple coils as a future release.)
Operating Hours / Day	F13	hours	With the two below, sets annual run hours.
Operating Days / Week	F14	days	
Operating Weeks / Year	F15	weeks	
Actual Primary Pump Head (Design)	F16	ft.	Head the pump works against at design flow (kPa in Metric).
Pump, VFD, Motor Efficiency (Average)	F17	%	Combined wire-to-water efficiency. If you do not know it, the help icon suggests 70–80%.
Design Delta T	F18	°F	What the coil was selected for.
Actual Delta T (Prior To Energy Valve Installation)	F19	°F	What it achieves today, before the Energy Valve – the "before" case.

Field	Cell	Unit	What it is
Future Delta T (Energy Valve Delta T Manager Setting)	F20	°F	What the Energy Valve's Delta-T Manager will hold – the “after” case.

The two Delta-T fields at the bottom are the whole estimate. Actual and Future are the before-and-after the entire comparison is built from, and the tool enforces a minimum gap of 2 °F between them (Future must be at least 2 degrees above Actual) so the pair always makes physical sense; a value outside the allowed range is refused when you type it.

Delta-Ts snap to six fixed scenarios: 4, 6, 8, 10, 12 and 14 °F

The Results tab compares six fixed Delta-T scenarios: **4, 6, 8, 10, 12 and 14 °F** (in Metric the same six steps, for example 3.33 for 6 °F). Actual and Future must be one of them. If you type another value, the tool adjusts it to the nearest scenario and tells you: 7 becomes 8, 13 becomes 14, and an Actual below 4 becomes 4. Actual can be at most 12 and Future at least 6, and Future is always at least 2 degrees above Actual. The adjusted values are what appear in the box and on the Results tab.

3.2 Steps 3–4: Operating time and climate region

Annual Operating Hours: 5497 hours
Climate Region: North East

Point	Bin	Load	% of Operating Time	Operating Hours
A	Design	100%	25.0%	1374
B	Peak	75%	50.0%	2748
C	Low	50%	25.0%	1374
D	Min	25%	0.0%	0
			100%	5497

Climate Region Degree Day Input Data

Climate Region	Cooling Degree Days	Humidity Adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.00
South	2004	30%	1.00
South West	2750	5%	1.00
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.00
Custom Region	800	0%	0.52
Average US	1526	19%	1.00

Selected Value: 0.88

the drop down box

% of Operating Time
Enter Percentages of operating time.

Step 4

Verify your Climate Region or select custom and insert your own value.

U.S. Standard Regions

Figure 4 – Steps 3 and 4 – operating time, load points and climate region.

Annual operating hours are computed from your three entries and then adjusted for climate: hours per day × days per week × weeks per year × the **climate multiplier** for your region (the Selected Value beneath the region table). With the example values, $20 \times 6 \times 52 = 6,240$ hours × 0.88 (North East) = 5,497 hours. The multiplier is never more than 1.00: a hot region can reduce the hours the coil needs to run but never adds hours beyond the schedule you entered.

Rather than simulate every hour of the year, the tool uses four representative operating points and gives each a share of the running time. Only the shares for points A, B and C are editable; point D is calculated as whatever is left. Excel refuses an entry that would make $A + B + C$ more than 100%, so point D can never go negative.

Point	Cell	Load	Default share	Meaning
A – Design	E33	100%	25%	Full-load hours.
B – Peak	E34	75%	50%	The bulk of a typical year.
C – Low	E35	50%	25%	Light load.
D – Min	E36	25%	0%	Not editable – calculated as whatever is left after A, B and C (0% when they total 100%). Its hours are included in the annual total.

Climate Region (cell E30) is a drop-down. Each region supplies a cooling-degree-day figure and a humidity adjustment, from which the multiplier is calculated relative to the average US region and capped at 1.00 (the table shows the values the tool applies):

Region	Cooling degree days	Humidity adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.00
South	2004	30%	1.00
South West	2750	5%	1.00
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.00
Custom Region	800 (editable)	0% (editable)	0.52
Average US	1526	19%	1.00

For a site that fits none of the standard regions, type your own cooling degree days and humidity adjustment in the **Custom Region** row (cells E53 and F53), then choose “Custom Region” in the drop-down. The multiplier is (region degree days ÷ average degree days) × (1 + humidity adjustment), never more than 1.00. The degree days must be between 1 and 6,000.

3.3 Step 5: Financial inputs and results

Financial Inputs					Step 5
Investment Cost (Energy Valve + Installation)	\$4,000.00	\$			Enter the estimated cost to purchase Energy Valves and the cost of installation.
Discount Rate	4.0%	%			
Estimated Measure Life	15	years			
	5 years	10 years	15 years	20 years	
Net Present Value (NPV)	\$3,488	\$9,643	\$14,701	\$18,859	
Savings-Investment Ratio (SIR)	1.87	3.41	4.68	5.71	
Simple Payback Period	2.4 years				

Figure 5 – Step 5 – financial inputs and the three financial results at four horizons.

Field	Cell	Unit	What it is
Investment Cost (Energy Valve + Installation)	F62	\$	Use the incremental cost over the valve that would otherwise have been fitted, not the whole valve package.
Discount Rate	F63	%	Converts future dollars into today's. Usually the organization's cost of capital or required return. 3–6% is common for public and utility-funded work.
Estimated Measure Life	F64	years	How long the measure is expected to keep saving. Feeds the lifetime figures on the Results tab.

Below the inputs, three results appear at 5, 10, 15 and 20 years: the **Net Present Value (NPV)**, the **Savings-Investment Ratio (SIR)** and the **Simple Payback Period**. All three are explained in Section 7. The small (i) icons beside the Discount Rate, NPV and SIR open a plain-language explanation of each.

3.4 The help icons and the Disclaimer

Every round blue (i) icon opens a short help message in your chosen language (English, Español, Português or Français). Nearly every input and result has one. On the Home tab they explain each design input (Electricity Cost, Design Coil Load, the operating hours, Pump Head, Pump Efficiency and the three Delta-Ts), Annual Operating Hours, the Climate Region and its load-point table, the three climate columns (Cooling Degree Days, Humidity Adjustment and Multiplier), the Custom Region row, the Selected Value, and the financial inputs and results (Investment Cost, Discount Rate, Estimated Measure Life, NPV, SIR and Simple Payback). The Pump Savings, Results and Charts tabs have their own (Sections 4, 6 and 5). Click Close to dismiss.

A **Disclaimer** band at the foot of the Home tab states that the tool is not copyrighted by Belimo Aircontrols (USA), Inc., is strictly an internal estimating tool, that results depend on the accuracy of the data entered and do not consider external factors, and that it looks at the effects of pump costs over various Delta-Ts for a single coil using standard pump affinity rules.

4. The Pump Savings Tab

The engine. Nothing here needs filling in except one factor – but it is worth understanding what it is doing.

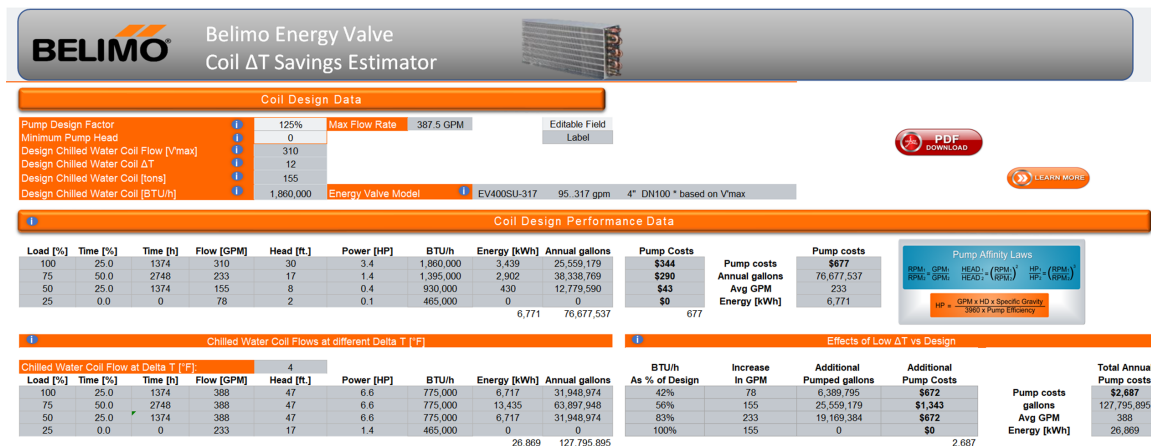


Figure 6 – The Pump Savings tab: design data, valve selection, the design-point performance table and the first Delta-T block.

4.1 Design data

Field	Cell	What it is
Pump Design Factor	F5	The only editable cell on this tab (default 125%). A safety margin on design flow – how much bigger the pump is than the coil strictly needs. It caps how much flow the pump can deliver (the Max Flow Rate beside it is design flow × this factor). Leave it alone unless you know the real oversizing.
Minimum Pump Head	F6	The lowest head the pump can fall to at part load. It is part of the calculation (head follows the flow squared but never drops below this value), but in this release it is locked and fixed at 0 (no floor); entering a value is a future feature.
Design Chilled Water Coil Flow [V'max]	F7	Your Design Coil Load from the Home tab.
Design Chilled Water Coil ΔT	F8	Your Design Delta T.
Design Chilled Water Coil [tons] / [BTU/h]	F9, F10	The design cooling capacity implied by the flow and Delta-T (tons = GPM × ΔT ÷ 24; BTU/h = GPM × ΔT × 500).

4.2 Energy Valve selection

Using the design flow, the tool picks the Energy Valve model that covers it and shows the model, its flow range and its pipe size beside the design data (for example EV400SU-317, 95–317 gpm, 4" DN100). The choice is based on V'max, the design flow. The range is shown in whichever units you selected.

Design flow (GPM) from	Model	Flow range	Pipe size
1.65	EV050	1.65–6.6 gpm	0.5" DN15
6.6	EV075	2.7–11 gpm	0.75" DN20
12	EV100	4.5–18.2 gpm	1" DN25
19	EV125	7.1–28.5 gpm	1.25" DN32
29	EV150	11–44 gpm	1.5" DN40
45	EV200	16.5–66 gpm	2" DN50
67	EV200H	25–100 gpm	2" DN50
101	EV250SU-127	38–127 gpm	2.5" DN65
128	EV300SU-180	54–180 gpm	3" DN80
181	EV400SU-317	95–317 gpm	4" DN100
318	EV500SU-495	149–495 gpm	5" DN125
496	EV600SU-713	214–713 gpm	6" DN150
714	Dual EV's	Up to 1400 gpm	–

The design flow is the **Design Coil Load** you enter on the Home tab, so changing it changes the model, the flow range and every energy, water and cost figure on this tab and on the Results tab. A flow below 1.65 GPM shows "Below EV range"; a flow of 714 GPM or more shows "Dual EV's".

The selection is a guide for the conversation, not a sizing calculation; confirm the valve with a full sizing before quoting.

4.3 The performance blocks

Below the design data, the tab has a block for the design case and then one block for each Delta-T scenario (4, 6, 8, 10, 12 and 14 °F). Each block carries the four load points across flow, head, power, energy and cost, and they all work the same way: for each load point the tool works out the flow needed at that Delta-T, the head the pump sees at that flow, the resulting pump power and, from the hours in that bin, the energy, water volume and cost. Add the four rows and you have that Delta-T's annual pumping bill. Section 8 gives the formulas.

4.4 Saving the tab as a PDF

The picture button at the top right of the tab saves the sheet you are on as a PDF. It offers a file name that includes the sheet name and the date and time, in the workbook's folder, through a Save As dialog, and confirms with the file location when done. The sheet prints on two landscape pages; the valve lookup table to its right is not included.

Help icons on this tab

The (i) beside Pump Design Factor explains it as the percentage of additional pumping available due to over-sizing or future design. The (i) beside Minimum Pump Head explains the pressure floor and notes that its value is fixed at 0 in this release. Each design-data line (flow, Delta T, tons, BTU/h) has an (i) showing how it is derived, the (i) beside the Energy Valve model explains how the model is chosen, and the (i) on the performance and Delta-T section bars explain how to read those tables.

5. Flow Curves and Charts

Reference material. Nothing to fill in on either tab. These tabs are there for the technical conversation – how a control signal maps to valve position and flow, and what that looks like plotted. They are useful when a customer or controls contractor asks how the valve behaves across its range; they are **not** part of the savings calculation.

5.1 Flow Curves

Flow Curves relates the DDC control signal to the actual flow through the valve, at the design flow you entered. Down the left, each row is one step of valve opening (0–100%) with the corresponding control voltage; the columns show the flow for a linear characteristic and for an equal-percentage characteristic, and the heat transfer (BTU/h) at the design Delta-T. The blue columns are the ones to use, and a worked example is printed at the right: an Energy Valve with a V'max of 310 GPM at 2.88 volts flows 17.67 GPM at 6% valve opening, producing 106,023 BTU/h at a 12 °F Delta-T (this example text is fixed and does not change with your inputs). The right-hand block feeds the charts.

BELIMO® Belimo Energy Valve									
Used to Calculate Projected Flow									
DDC Control signal		Valve Position & Flow based on DDC Voltage							
Voltage	Voltage	Linear Part	Equal Part	Vmax	Vmax	ΔT	btu/h		
0.50	2.00	Linear	EQ %	Linear	EQ %				
10.00	10.00			310	310	12	btu/h		
In %	In V	In V	In %	In %	g/min	g/min	°F	btu/h	
0	0.50	2.00	0.0%	4.0%	0.00	0.00		0	
1	0.60	2.08	0.4%	4.1%	0.00	0.00		0	
2	0.69	2.16	0.7%	4.3%	0.00	0.00		0	
3	0.79	2.24	1.1%	4.4%	0.00	0.00		0	
4	0.88	2.32	1.4%	4.5%	0.00	0.00		0	
5	0.98	2.40	1.8%	4.7%	0.00	0.00		0	
6	1.07	2.48	2.1%	4.9%	0.00	0.00		0	
7	1.17	2.56	2.5%	5.0%	0.00	15.54		93,210	
8	1.26	2.64	2.8%	5.2%	8.68	16.04		96,260	
9	1.36	2.72	3.2%	5.3%	9.77	16.57		99,410	
10	1.45	2.80	3.5%	5.5%	10.85	17.11		102,663	
11	1.55	2.88	3.9%	5.7%	11.94	17.67		106,023	
12	1.64	2.96	4.2%	5.9%	13.02	18.25		109,492	
13	1.74	3.04	4.6%	6.1%	14.11	18.85		113,076	
14	1.83	3.12	4.9%	6.3%	15.19	19.46		116,776	
15	1.93	3.20	5.3%	6.5%	16.28	20.10		120,597	
16	2.02	3.28	5.6%	6.7%	17.36	20.76		124,544	
17	2.12	3.36	6.0%	6.9%	18.45	21.44		128,619	
18	2.21	3.44	6.3%	7.1%	19.53	22.14		132,828	
19	2.31	3.52	6.7%	7.4%	20.62	22.86		137,175	
20	2.40	3.60	7.0%	7.6%	21.70	23.61		141,664	
21	2.50	3.68	7.4%	7.9%	22.79	24.38		146,300	
22	2.59	3.76	7.7%	8.1%	23.87	25.18		151,087	
23	2.69	3.84	8.1%	8.4%	24.96	26.01		156,031	
24	2.78	3.92	8.4%	8.7%	26.04	26.86		161,137	
25	2.88	4.00	8.8%	8.9%	27.13	27.74		166,410	
26	2.97	4.08	9.1%	9.2%	28.21	28.64		171,856	
27	3.07	4.16	9.5%	9.5%	29.30	29.58		177,480	
28	3.16	4.24	9.8%	9.9%	30.38	30.55		183,287	
29	3.26	4.32	10.2%	10.2%	31.47	31.55		189,285	
30	3.35	4.40	10.5%	10.5%	32.55	32.58		195,480	
31	3.45	4.48	10.9%	10.9%	33.64	33.65		201,876	
32	3.54	4.56	11.2%	11.2%	34.72	34.75		208,483	
33	3.64	4.64	11.6%	11.6%	35.81	35.88		215,305	
34	3.73	4.72	11.9%	12.0%	36.89	37.06		222,351	
35	3.83	4.80	12.3%	12.3%	37.98	38.27		229,627	
36	3.92	4.88	12.6%	12.7%	39.06	39.52		237,141	
37	4.02	4.96	13.0%	13.2%	40.15	40.82		244,901	
38	4.11	5.04	13.3%	13.6%	41.23	42.15		252,916	
39	4.21	5.12	13.7%	14.0%	42.32	43.53		261,192	
40	4.30	5.20	14.0%	14.5%	43.40	44.96		269,739	
41	4.40	5.28	14.4%	15.0%	44.49	46.43		278,566	
42	4.49	5.36	14.7%	15.5%	45.57	47.95		287,682	
43	4.59	5.44	15.1%	16.0%	46.66	49.52		297,096	
44	4.68	5.52	15.4%	16.5%	47.74	51.14		306,818	
45	4.78	5.60	15.8%	17.0%	48.83	52.81		316,859	
46	4.87	5.68	16.1%	17.6%	49.91	54.54		327,228	
47	4.97	5.76	16.5%	18.2%	51.00	56.32		337,936	
48	5.06	5.84	16.8%	18.8%	52.08	58.17		348,994	
49	5.16	5.92	17.2%	19.4%	53.17	60.07		360,415	
50	5.25	6.00	17.5%	20.0%	54.25	62.03		372,209	

Used for Charts									
Equal Part	Vmax	Y	Vmax	btu/h					
EQ %	EQ %	2.00	EQ%						
10.00	10.00		310	btu/h					
In %	In V	In V	g/min	btu/h					
4.0%	0.00	2.00	0.00	0					
4.1%	0.00	2.08	0.00	0					
4.3%	0.00	2.16	0.00	0					
4.4%	0.00	2.24	0.00	0					
4.5%	0.00	2.32	0.00	0					
4.7%	0.00	2.40	0.00	0					
4.9%	0.00	2.48	0.00	0					
5.0%	15.54	2.56	15.54	93,210					
5.2%	16.04	2.64	16.04	96,260					
5.3%	16.57	2.72	16.57	99,410					
5.5%	17.11	2.80	17.11	102,663					
5.7%	17.67	2.88	17.67	106,023					
5.9%	18.25	2.96	18.25	109,492					
6.1%	18.85	3.04	18.85	113,076					
6.3%	19.46	3.12	19.46	116,776					
6.5%	20.10	3.20	20.10	120,597					
6.7%	20.76	3.28	20.76	124,544					
6.9%	21.44	3.36	21.44	128,619					
7.1%	22.14	3.44	22.14	132,828					
7.4%	22.86	3.52	22.86	137,175					
7.6%	23.61	3.60	23.61	141,664					
7.9%	24.38	3.68	24.38	146,300					
8.1%	25.18	3.76	25.18	151,087					
8.4%	26.01	3.84	26.01	156,031					
8.7%	26.86	3.92	26.86	161,137					
8.9%	27.74	4.00	27.74	166,410					
9.2%	28.64	4.08	28.64	171,856					
9.5%	29.58	4.16	29.58	177,480					
9.9%	30.55	4.24	30.55	183,287					
10.2%	31.55	4.32	31.55	189,285					
10.5%	32.58	4.40	32.58	195,480					
10.9%	33.65	4.48	33.65	201,876					
11.2%	34.75	4.56	34.75	208,483					
11.6%	35.88	4.64	35.88	215,305					
12.0%	37.06	4.72	37.06	222,351					
12.3%	38.27	4.80	38.27	229,627					
12.7%	39.52	4.88	39.52	237,141					
13.2%	40.82	4.96	40.82	244,901					
13.6%	42.15	5.04	42.15	252,916					
14.0%	43.53	5.12	43.53	261,192					
14.5%	44.96	5.20	44.96	269,739					
15.0%	46.43	5.28	46.43	278,566					
15.5%	47.95	5.36	47.95	287,682					
16.0%	49.52	5.44	49.52	297,096					
16.5%	51.14	5.52	51.14	306,818					
17.0%	52.81	5.60	52.81	316,859					
17.6%	54.54	5.68	54.54	327,228					
18.2%	56.32	5.76	56.32	337,936					
18.8%	58.17	5.84	58.17	348,994					
19.4%	60.07	5.92	60.07	360,415					
20.0%	62.03	6.00	62.03	372,209					

Projected Flow Calculations

Example:
 An Energy Valve with a V'max of 310 GPM
 At 2.88 volts, you would be flowing 17.67 GPM
 At 6% Valve Opening
 Producing 106,023 btu's at a 12 DeltaT

Hint: Refer to the Blue Columns

Figure 7 – The top of the Flow Curves tab.

5.2 Charts

The same relationships plotted: percent of valve open to GPM, DDC voltage to GPM, DDC voltage to GPM and BTU/h together, and GPM to BTU/h (a straight line, because heat transfer is flow × Delta-T × 500 at a fixed Delta-T).

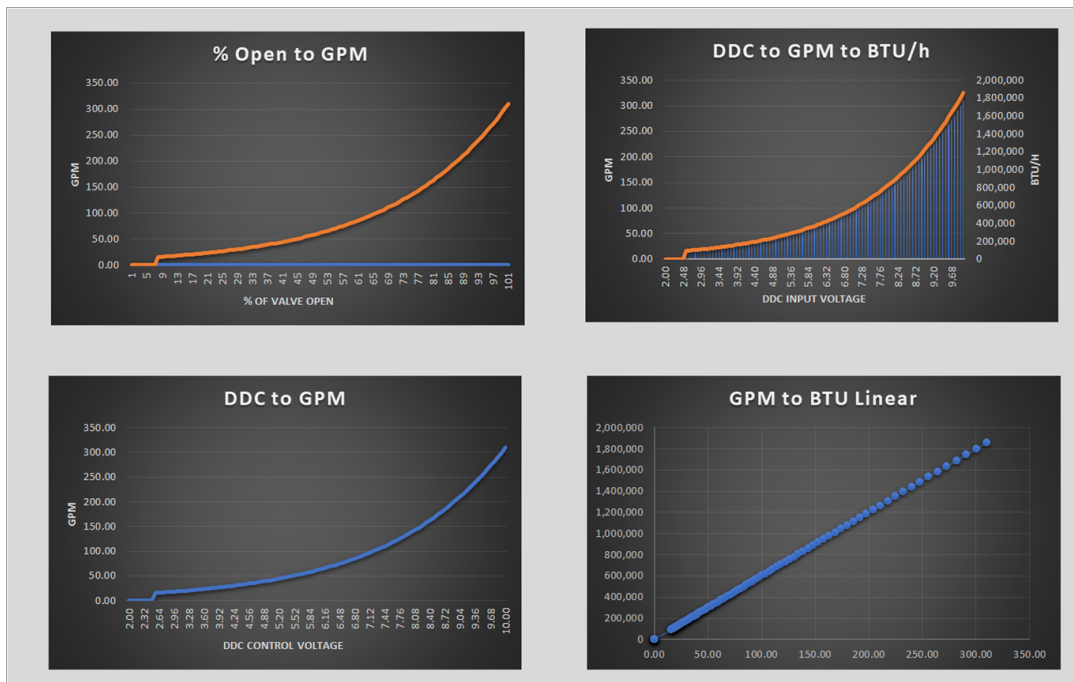


Figure 8 – The Charts tab.

6. The Results Tab

The customer-facing page. Nothing to fill in – this is what you present or print. Before and after, the full Delta-T range color-graded from worst to best, and an interactive slider beneath.

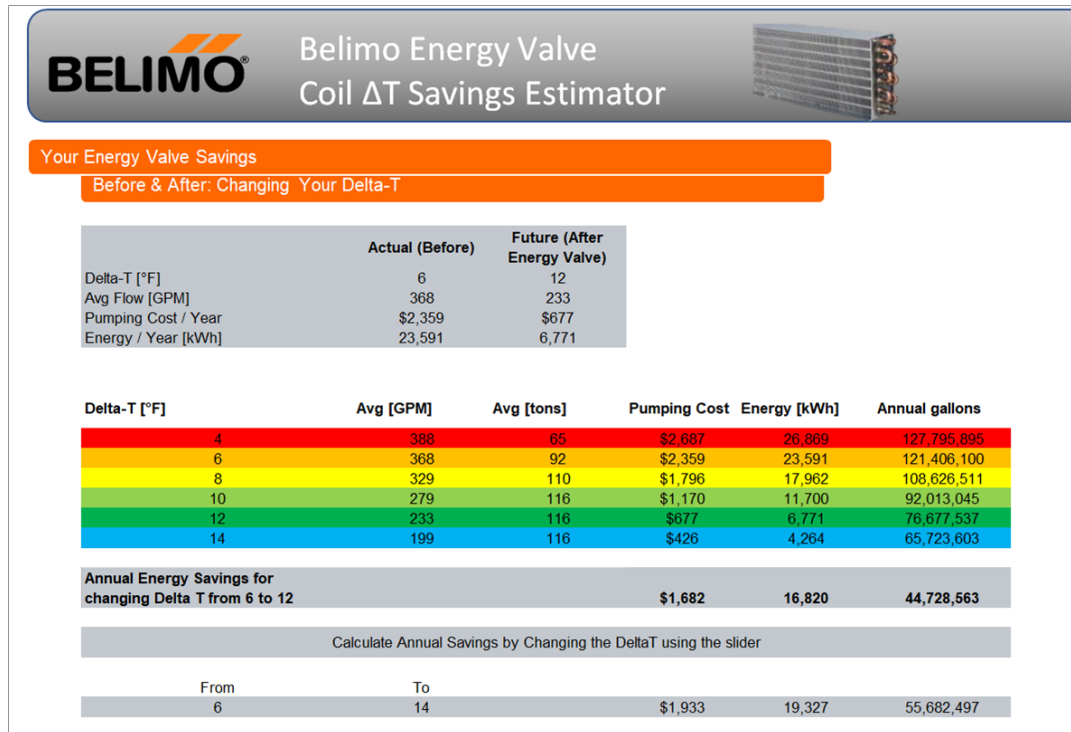


Figure 9 – The top of the Results tab: before and after, the Delta-T scenarios and the slider.

6.1 What each block shows

Block	Contents
Before & After	Your Actual and Future Delta-T side by side: Delta-T, average flow, pumping cost per year and energy per year.
Delta-T table	Six scenarios (4–14 $^{\circ}$ F) color-graded red (worst) to blue (best), with average flow, average tonnage, pumping cost, energy and annual water volume.
Annual Energy Savings	The headline line: dollars, kilowatt-hours and gallons saved by moving from your Actual Delta-T to your Future one.
The slider	Set any “To” Delta-T with the spin buttons and the savings line beneath recalculates live, moving 2 degrees at a time. The “From” case stays pinned to the coil’s Actual Delta-T. The slider runs from one step below the Actual Delta-T up to one step above the Design Delta-T, but never outside the six scenarios (4–14 $^{\circ}$ F), so every click lands on a scenario in the table above.

The slider is the most persuasive thing on the tab: it answers “what if we only get halfway there?” while the customer watches, instead of forcing a single optimistic number. Its (i) icon explains that the slider changes the Delta-T savings by 2 degrees at a time.

6.2 Greenhouse gas equivalents and financial summary

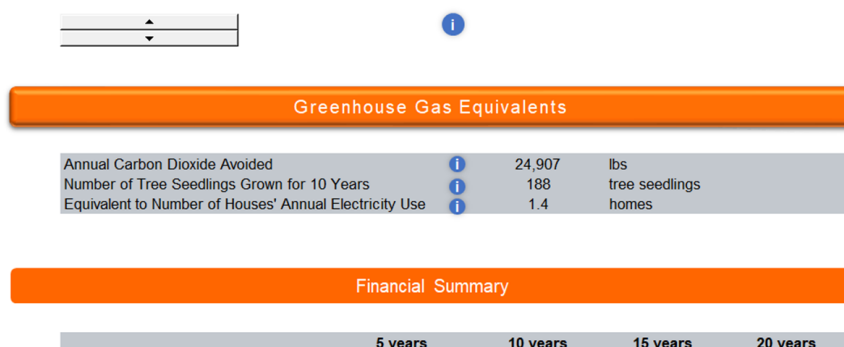


Figure 10 – The slider control, greenhouse gas equivalents and financial summary.

The greenhouse-gas block expresses the saving as pounds of carbon dioxide avoided and as two equivalents people can picture – tree seedlings grown for ten years, and a number of homes' annual electricity use. The (i) icons note that the calculations follow EPA methods; the carbon dioxide is the energy saved $\times$ 1.4808 lb per kWh (the U.S. EPA national non-baseload rate, eGRID 2022, including transmission and distribution losses), a tree seedling grown for ten years absorbs 0.060 metric tons of CO₂, and the homes figure is the energy saved divided by 12,194 kWh, the annual electricity use of a home (EPA). The financial summary repeats the three results from the Home tab at 5, 10, 15 and 20 years.

6.3 Utility rebate support

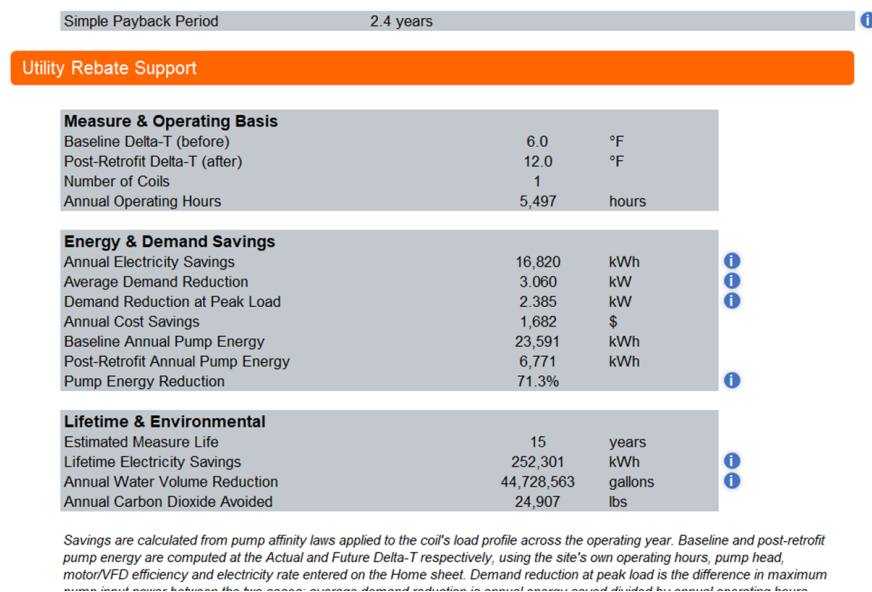


Figure 11 – The Utility Rebate Support block.

This block restates the result in the shape a utility program wants: the measure basis (baseline and post-retrofit Delta-T, coil count, annual hours), the energy and demand savings, and the lifetime and environmental figures. It closes with a methodology note, so a reviewer can see the calculation basis without having to ask.

Two demand figures, and they are not the same thing

Average Demand Reduction is the annual energy saving divided by annual hours. **Demand Reduction at Peak Load** is the difference in pump power at the design point. A rebate form usually wants the peak figure; do not quote one where the other is asked for.

7. The Financial Results

Three numbers, shown at four horizons. These are the ones that get misread, so they are worth a careful read.

7.1 First, the idea behind all three: present value

Money in the future is worth less than money now – you could have invested today’s dollar in the meantime. “Present value” is what a future amount is worth today, once you have discounted it back at the rate you set in Step 5. At a 4% discount rate, a dollar saved ten years from now counts as about 68 cents today. The discount rate is a lever: raise it and future savings count for less, so both Net Present Value and the Savings-Investment Ratio get smaller. It is not a fudge factor – set it to what your organization actually requires and leave it there.

7.2 Net Present Value (NPV)

Take every year’s saving, discount it back to today, add them up, and subtract what you spent on day one. What is left is the value the project adds, expressed in today’s dollars. **Above zero** means worth doing; **below zero** means not at this discount rate – which does not mean the project loses money, only that it does not clear the return you demanded of it. It is shown at 5, 10, 15 and 20 years and grows with the horizon, because the spend happens once at the start while the savings keep accruing.

7.3 Savings-Investment Ratio (SIR)

The present value of everything you get back, divided by what you put in. Because it is a ratio rather than an amount, it lets you compare projects of very different sizes on equal terms. **Above 1.0** returns more than it costs; **below 1.0** does not pay back in present-value terms. An SIR of 1.9 means every dollar invested returns about \$1.90 in today’s money. Many public-sector and utility incentive programs set a minimum ratio as an eligibility condition, which is why it sits beside NPV on both the Home and Results tabs.

7.4 Simple Payback Period

How long the savings take to repay the investment, ignoring the time value of money entirely: the investment divided by the annual cost saving, shown in years. It is the number a building owner asks for first, and the easiest to explain. Read it as a sanity check, not as the whole case – it ignores everything that happens after payback, so it systematically undersells a long-lived measure. Quote it because people ask for it, then show the Net Present Value.

Which one do you lead with?

For a building owner, Simple Payback then Net Present Value. For a finance department, Net Present Value. For a utility or public-sector program, the Savings-Investment Ratio – it is often written straight into the eligibility rules.

7.5 Reading the four horizons

Each result is repeated at 5, 10, 15 and 20 years, answering the same question over a different holding period: if we judged this on an N-year view, how would it look? Short horizons always look worse, because the money goes out on day one while the savings accumulate. Quote the horizon that matches how the customer actually evaluates capital – and say which horizon you are quoting.

7.6 Worked example (the values the workbook opens with)

Item	Value
Coil	310 GPM, 12 °F design Delta-T (155 tons), 30 ft head, 70% efficiency, electricity \$0.10/kWh
Operating time	20 h/day, 6 days/week, 52 weeks/year, North East (multiplier 0.88) = 5,497 hours per year
Actual → Future Delta-T	6 °F → 12 °F
Pumping cost before / after	\$2,359 per year → \$677 per year (23,591 kWh → 6,771 kWh, a 71.3% reduction)
Annual saving	\$1,682 and 16,820 kWh, and 44,728,563 gallons less water pumped
Investment / discount rate / life	\$4,000 / 4% / 15 years
NPV at 5 / 10 / 15 / 20 years	\$3,488 / \$9,643 / \$14,701 / \$18,859
SIR at 5 / 10 / 15 / 20 years	1.87 / 3.41 / 4.68 / 5.71
Simple payback	2.4 years

8. How the Numbers Are Calculated

For anyone who wants to check the arithmetic. All calculations are done in US units and converted for display.

Quantity	Calculation
Design tonnage	Design flow (GPM) × Design Delta-T ÷ 24
Flow needed at a load point and Delta-T	Design tonnage × 24 × (load %) ÷ Delta-T, limited to the pump's maximum flow (design flow × Pump Design Factor). In the design case the flow is simply load % × design flow.
Pump head at that flow	Design head × (flow ÷ design flow) ² (the pump affinity law; never below Minimum Pump head)
Pump power (HP)	Flow (GPM) × head (ft) ÷ (3960 × efficiency)
Energy (kWh)	Hours in the bin × HP × 0.746
Cost (\$)	kWh × electricity cost
Annual water volume (gal)	Flow × 60 × hours in the bin
Annual operating hours	Hours/day × days/week × weeks/year × climate multiplier (never more than 1.00)
Climate multiplier	(Region degree days ÷ average degree days) × (1 + region humidity adjustment), capped at 1.00
Annual saving	Annual pumping cost (or kWh, or gallons) at Actual Delta-T minus the same at Future Delta-T
Average demand reduction (kW)	Annual kWh saved ÷ annual operating hours
NPV (N years)	– Investment + annual cost saving × $[1 - (1 + r)^{-N}] ÷ r$ (r = discount rate; N × saving if r = 0)
SIR (N years)	Annual cost saving × $[1 - (1 + r)^{-N}] ÷ r ÷$ Investment
Simple payback	Investment ÷ annual cost saving
CO₂ avoided (lb)	kWh saved × 1.4808 (U.S. EPA national non-baseload rate, eGRID 2022, with transmission and distribution losses)
Tree seedlings	CO ₂ avoided converted to metric tons ÷ 0.060 tons per seedling grown for 10 years (U.S. EPA)
Homes	kWh saved ÷ 12,194 kWh per home per year (U.S. EPA)

The four load points (Load 100/75/50/25%) and their shares of the year come from Steps 3 and 4. The pump is assumed to be a single pump sized to meet and exceed the coil's design flow by the Pump Design Factor, as stated on the About screen.

9. Practicalities and Troubleshooting

9.1 Presenting and printing

The Results tab is the page to show. Print or export it once the inputs are settled – it carries the comparison, the savings, the financial summary and the rebate block. Every tab has a print area, so File › Print or Export gives Results, Pump Savings and Home on two landscape pages each, Charts on one page and Flow Curves on two. To save a sheet as a PDF from the Pump Savings tab, use its PDF button (Section 4.4); on other tabs use Excel's File › Export or Print.

9.2 Language

Switching language on the Home tab re-labels the whole workbook, including the Results tab headings, the help messages and the chart legends. Numbers and units follow the separate Units drop-down. Your language choice is applied each time the workbook opens.

9.3 If something looks wrong

Symptom	Check
A cell will not let you type.	It is a calculated result. Compare it against the shading key in Section 2.4.
Savings look implausibly large.	Check the gap between Actual and Future Delta-T, and the Pump Design Factor on the Pump Savings tab.
The financial results look wrong.	Almost always the investment figure. Confirm it holds the incremental cost of the Energy Valve over the valve that would have been fitted anyway.
Delta-T entries are rejected.	The tool enforces a minimum gap of 2 degrees between Actual and Future, and a sensible range on Design. Adjust the pair together.
A box says "The results compare six fixed Delta T scenarios".	You typed an Actual or Future Delta-T that is not one of the six scenarios (4, 6, 8, 10, 12, 14 °F). The tool adjusted it to the nearest one and the Home and Results tabs use that value (Section 3.1).
The operating-time total shows an error.	The shares for points A, B and C in Step 3 add up to more than 100%. Reduce one of them.
Numbers did not change after an edit.	Click away from the cell to commit the entry, or press F9 to recalculate.
Buttons and the About screen do nothing.	Macros are disabled. Click Enable Editing / Enable Content (Section 2.1).
Check for Updates reports a problem.	The update source could not be reached. The tool keeps working; try again later.

This tool produces an estimate. Results depend on the accuracy of the inputs and on conditions specific to the installation. Questions, corrections and feature requests go to the maintainer named on the About screen.

Appendix. Release History

Release	Date	Changes
4.5	2026-10-05	Fixed: Demand Reduction at Peak Load on the Results tab was wrong in Metric units. Annual operating hours can no longer exceed the schedule you enter: the climate multiplier is capped at 1.00 (South East, South, South West, West and Average US were above it, up to 1.93). Carbon dioxide, tree-seedling and homes equivalents use the current U.S. EPA factors, as in the Savings Estimator (1.4808 lb per kWh, 0.060 t per seedling, 12,194 kWh per home). The load-point shares can no longer add up to more than 100%, and custom cooling degree days must be 1 to 6,000. Every tab now has a print area: Results, Pump Savings and Home print on two landscape pages and the Pump Savings PDF no longer includes the valve lookup table. Spanish, Portuguese and French: shorter labels where text ran into values or under (i) icons, wider Pump Savings columns, larger Step boxes on Home (the Step 1 text is shorter and Step 2 says Flow instead of GPM), a translated Results banner, and the valve table names “Dual EV’s” and “Up to” are translated. Removed leftover “IPLV” prompts on two Home inputs.
4.4	2026-09-29	Fixed: the Design Coil Load on the Home tab now drives the Pump Savings and Results tabs and the Energy Valve model choice; before, it only changed the Flow Curves tab, so savings were always shown for a 310 GPM coil. Fixed: the Delta-T slider on the Results tab could stop changing the savings (a Design Delta-T above 12 °F left its upper clicks on the 14 °F figures) or show #N/A (an Actual Delta-T of 4 °F); it now stays within the six scenarios and re-ranges when the Delta-T inputs on the Home tab change. Help icons: 44 new (i) icons on the Home, Pump Savings, Results and Charts tabs explain nearly every input and result in English, Spanish, Portuguese and French. The Minimum Pump Head help was corrected (the value is part of the calculation but is fixed at 0 in this release). Fixed typos in the help, Step 2 and Disclaimer text and in labels (“DeltaT”, “btuH” and “btu’s” now read Delta T and BTU/h); the Spanish NPV help now says VPN, like its label.
4.3	2026-09-24	When a newer release is published, the tool now tells you when you open it, once for each new version (Yes installs it, No carries on). Check for Updates on the About screen is always available. Fixed: on a PC whose regional settings use a comma for decimals (for example Spanish, Portuguese or French), Check for Updates could keep offering the version you already had.
4.2	2026-09-24	Added the standard admin keys used in every Belimo tool: F12 asks for the admin password and then shows and unprotects every sheet; F11 puts every sheet back exactly as it was. One password now protects every sheet in every tool. These keys are for maintainers and do not change how you use the tool. Corrected three spelling mistakes. Operating Hours / Day and Operating Days / Week no longer accept 0 (it produced #DIV/0! errors on the Pump Savings sheet). The workbook’s file name no longer ends in _ML (Belimo Energy Valve Coil Estimator.xlsm); automatic updates fetch it under the new name.
4.1	2026-09-23	Actual and Future Delta-T are now matched to the six scenarios the results use (an entry such as 7 or 13 is adjusted and you are told; an Actual below 4 no longer shows errors). The Min load point now counts its hours. The homes-equivalent figure uses the 12,258 kWh per home stated in its help text. Removed a stray period from the savings heading on the Results tab.

Release	Date	Changes
4.0	2026-09-22	Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen's version line is now live. The About screen also has a Download User Manual button, which fetches this manual.
Earlier	–	Multi-language (English, Español, Português, Français) and US/Metric units; the Home tab reorganized around a Step 1 for language and units; the consolidated Results tab with the slider, greenhouse-gas equivalents, financial summary and Utility Rebate Support block.